

Expanded Machine Learning Algorithm Family Map

1. Distance / Instance-Based Learning

- k-Nearest Neighbors (k-NN)
- Weighted k-NN
- Radius Neighbors
- Nearest Centroid
- Locally Weighted Regression (LWR)
- Case-Based Reasoning

2. Tree-Based Methods (Non-Ensemble)

- ID3
- C4.5
- C5.0
- CART (Classification and Regression Trees)
- CHAID
- QUEST
- M5 Model Tree
- Conditional Inference Trees (CTree)

3. Bagging-Based Ensembles

- Bagging
- Random Forest
- Extra Trees (Extremely Randomized Trees)
- Random Subspace Method
- Random Patches

4. Boosting-Based Ensembles

- AdaBoost
- Gradient Boosting Machines (GBM)
- Gradient Boosted Decision Trees (GBDT)
- XGBoost
- LightGBM
- CatBoost
- LogitBoost
- BrownBoost
- LPBoost

5. Meta-Ensembles

- Voting Classifier (Hard Voting)
- Voting Classifier (Soft Voting)
- Voting Regressor
- Stacking (Stacked Generalization)
- Blending
- Super Learner
- Cascading Ensembles

6. Performance Evaluation

Classification Metrics:

- Confusion Matrix
- Accuracy
- Precision
- Recall (Sensitivity)
- Specificity
- F1-Score

- F-beta Score
- ROC Curve
- AUC-ROC
- Precision-Recall Curve
- Log Loss (Cross-Entropy Loss)
- Matthews Correlation Coefficient (MCC)
- Cohen's Kappa

Regression Metrics:

- MAE
- MSE
- RMSE
- RMSLE
- R^2 Score
- Adjusted R^2
- MAPE

Validation Methods:

- Train/Test Split
- Hold-Out Validation
- K-Fold Cross Validation
- Stratified K-Fold
- Repeated K-Fold
- Leave-One-Out (LOOCV)
- Leave-P-Out
- Monte Carlo Cross Validation
- Time Series Cross Validation (Walk Forward Validation)
- Nested Cross Validation

Related Tree-Based Specialized Algorithms

- Isolation Forest
- Random Survival Forest
- Rotation Forest
- Oblique Decision Trees
- Hoeffding Trees (Streaming Data)